Executive Summary

Factory-to-Customer Shipping Route Efficiency · Nassau Candy Distributor

Prepared for government and public-sector stakeholders · Based on 10,194 shipments, Jan 2024 – Dec 2025

Purpose

Nassau Candy distributes confectionery nationally from five factories to customers in 59 US states and Canadian provinces. This review measured how quickly each factory-to-customer route ships orders, where delays concentrate, and how service level affects speed, so that logistics decisions rest on evidence.

Headline figures

Measure	Value
Average shipping lead time	4.3 days
Shipments over 5 days	25%
Change in lead time, 2024 → 2025	+0.38 days (delays 18% → 29%)
Fastest vs slowest service	Same Day 0.4 d · Standard 5.3 d
Spread between fastest and slowest routes	1.7 days
Share of delays from Standard Class	94%

What we found

- **Delivery speed is set by service level, not distance.** Customers far from a factory are served as quickly as nearby ones; the ship mode chosen makes the difference.
- **Performance slipped across the network in 2025.** Every service level got slower, which points to order-processing capacity at the main factories rather than any single route.
- **Standard shipping is the main source of delay.** It carries 60% of shipments and 94% of late ones, with a wide 3–8 day window.
- **A few high-volume states are bottlenecks.** New York, Washington, Illinois, Florida, Michigan combine large customer bases with above-average delays.
- **Records needed correction.** Ship dates were stored with a year offset; corrected before analysis. Fixing this at source is a governance priority.

Recommended actions

1. Correct and validate ship-date recording so performance can be reported accurately.
2. Review dispatch capacity at the two main chocolate factories ahead of the September and holiday peaks.
3. Adopt a published ≤ 5-day standard-shipping commitment and monitor delay frequency monthly.
4. Work with carriers on the bottleneck states (New York, Washington, Illinois) to cut late shipments.
5. Capture freight cost and delivery dates to support cost-versus-speed decisions.

Expected benefit

Modelled with the dashboard's scenario simulator on the full 2024–2025 shipment base:

Scenario	Avg lead time (d)	Delay %	Late shipments avoided
Baseline	4.29	24.7%	0
Upgrade 25% of Standard to Second	4.03	20.0%	481
Upgrade 25% of Standard to First	3.87	18.9%	593
Recover 2025 slowdown (100%)	4.01	17.6%	726
Recover 2025 + upgrade 25% to Second	3.76	13.4%	1,157

Restoring 2024 dispatch performance alone would avoid about 726 late shipments over the period at no extra freight cost. Combined with upgrading a quarter of Standard Class orders, the late-shipment rate falls to about 13%. The interactive dashboard delivered with this review lets managers test these levers and track results continuously.